Question # 1

Tick the correct option

1.	If x = 0 and y = 0 then $\theta =$						
a.	90	b.	-90	c.	0	d.	Undefined
2.	What is true about modulus of a complex number:						
a.	$\lvert z\rvert > 0$	b.	$\lvert z\rvert \geq 0$	c.	$\lvert z\rvert \in R$	d.	None of these
3.	Cartesian form of the complex number (5,275)						
a.	(-4.4,-1.257)	b.	(0.087,0.99)	c.	(-0.087,-0.99)	d.	(0.44,-4.98)

Question # 2

Answer the following

1.	Express the following complex numbers in polar form: $\frac{1}{2} + \frac{\sqrt{3}i}{2}$
2.	Express the following complex numbers in polar form: $\frac{1-i}{1+i}$
3.	Convert in catesian/rectangular form: $\lvert z\rvert = \frac{10}{3}, \arg(z) = -\frac{17\pi}{12}$
4.	If z1 and z2 are two complex numbers then show that : $Arg\left(\frac{z_1}{z_2}\right) = Arg\, z_1 - Arg\, z_2$
5.	Divide $z_1 = 6(\cos 150° + i \sin 150°)$ by $z_2 = 3(\cos 30° + i \sin 30°)$ and express in $x + iy$ form.
6.	Write the equation $\arg(\bar{z} - 2 + i) = \frac{2\pi}{3}$ in cartesian form, if $z = x + iy$.
7.	Solve the equation $\lvert z - 2i\rvert = \lvert \bar{z} + 2\rvert$ for $z = x + iy$.
8.	If $z = x + iy$ and $w = \frac{1 - iz}{z - i}$, show that $\lvert w\rvert = 1 \Rightarrow z$ is real.
9.	Divide $z_1 = 6(\cos 150° + i \sin 150°)$ by $z_2 = 3(\cos 30° + i \sin 30°)$ and express in $x + iy$ form.
10.	If z_1 and z_2 are different complex numbers with $\lvert z_2\rvert = 1$, find $\left\lvert \frac{z_2 - z_1}{1 - \bar{z}_1 z_2} \right\rvert$.

Question # 5

Long Questions

1.	If $z = x + iy$ and $\arg\left(\frac{\bar{z} - 1 + 2i}{\bar{z} + 1 - 2i}\right) = \frac{9\pi}{4}$, show that $x^2 + y^2 - 4x + 2y - 5 = 0$.
2.	Encrypt the word "CLASS" by adding the complex encryption key $k = -3 + 4i$. Then decrypt it back to the original word.
3.	[redacted] If the impedance of circuit is $11(\cos 55.35° + i \sin 55.35°)$ ohms at a voltage of $25(\cos 30° + i \sin$ [redacted] $)$ V, find the value of current in the circuit.

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